

BENJAMIN HAN

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SKILLS AND QUALIFICATIONS

Programming Languages: Python, MATLAB, R, SQL, Java, Dart, Visual Basic for Applications (VBA), HTML, CSS
AI/ML: PyTorch, TensorFlow, scikit-learn, RAG pipelines, synthetic query generation, prompt engineering, LLM evaluation
Engineering and Design Software: Fusion 360, AutoCAD, LabVIEW, COMSOL Multiphysics, Onshape, SolidWorks

EDUCATION

University of Washington Seattle, WA Master of Applied Bioengineering GPA: 3.94	University of Washington Seattle, WA GPA: 3.47 B.S. Bioengineering: Data Science Minor: Applied Mathematics
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TECHNICAL PROJECTS

WiseFind: Medical AI Platform for Neurotrauma Assessment *September 2025 – Present*
University of Washington | Seattle, WA

- Designing and validating a Python data processing pipeline for a Retrieval-Augmented Generation system for handling neurotrauma queries, securing \$55,000 in funding from ITHS and the UW Neurosurgery Department
- Developed a synthetic query generation system that expanded LLM evaluation coverage from 20 manually written prompts to 400+ clinician-style queries, simulating realistic usage conditions
- Ensured project continuity and onboarding efficiency for new project members by maintaining detailed documentation and standard operating procedures (SOPs) for ongoing AI platform development

WiseSpine: Image-Focused AI Agent for Spine MR/CT Analysis *November 2025 – Present*
University of Washington | Seattle, WA

- Developed a Python-based spine reconstruction pipeline for automated Cobb angle measurement using 481 spine X-ray images and three deep learning segmentation models
- Built a segmentation benchmarking system for medical imaging using spatial analysis and quantitative metrics
- Accelerated scalable AI infrastructure design by collaborating with AWS engineers on pipeline development

CareChorus: Patient-Centered Care Navigation App *March 2026 – June 2026*
University of Washington | Seattle, WA

- Validated an AI-driven mobile application designed for patient treatment navigation across healthcare systems
- Designed an evaluation pipeline in Dart to benchmark classification and data extraction of medical documents
- Supported product development by collaborating with stakeholders on feature design and implementation

WORK EXPERIENCE

Jr. Quality Engineer *September 2024 – Present*
Bio-Therapeutic | Seattle, WA

- Conducted routine surveillance audits to verify compliance with Bio-Therapeutic quality standards and SOPs
- Developed three interactive Tableau dashboards integrating 1,000+ purchasing orders and supply records, improving operational visibility and enabling data-driven quality reporting
- Reduced manual data entry by automating Excel/VBA-based customer service logging and reporting workflows
- Analyzed quality, procurement, and customer service data to identify trends, support continuous process improvement, and inform management decision-making

Graduate/Undergraduate Lead Teaching Assistant *January 2024 – June 2026*
UW Bioengineering Department | Seattle, WA

- Strengthened student understanding of MATLAB programming and bioengineering design concepts by working with instructors to develop course materials, revise online modules, and lead undergraduate lectures
- Supported academic success through holding 150+ office hours and mentoring eight student engineering teams

Engineering Services Intern *September 2024 – June 2025*
UW Facilities | Seattle, WA

- Streamlined data entry workflows by developing a Python script to automate logging and metadata extraction
- Organized and maintained large structured datasets (190k+ records) of University of Washington building schematics, safety documents, and engineering records to improve data consistency
- Established tagging procedures for metadata information support efficient document retrieval

SummerWorks Engineering Intern *June 2023 – August 2023*
City of Portland Bureau of Environmental Services | Portland, OR

- Improved accessibility of engineering records by designing a VBA-based tool managing 500+ pipe data entries
- Reduced manual engineering calculations by implementing Excel-based automation for pump curve analysis
- Streamlined documentation workflows by annotating pump station schematics for infrastructure records